

Lab 1:

MongoDB – CRUD Operations Demonstration (Practice and Self Study)

```
// Show databases
show dbs

// Use database
use myDB
db

// Create collection and insert documents
db.createCollection("Student")
db.Student.insert({RollNo:1, Age:21, Cont:9876, email:"antara.de9@gmail.com"})
db.Student.insert({RollNo:2, Age:22, Cont:9976, email:"anushka.de9@gmail.com"})
db.Student.insert({RollNo:3, Age:21, Cont:5576, email:"anubhav.de9@gmail.com"})
db.Student.insert({RollNo:4, Age:20, Cont:4476, email:"pani.de9@gmail.com"})
db.Student.insert({RollNo:10, Age:23, Cont:2276, email:"Abhinav@gmail.com"})

// Read all documents
db.Student.find()

// Update
db.Student.updateOne({"RollNo": 10}, {$set: {"email": "tarun.de9@gmail.com"}})

// Customers - aggregate
db.createCollection("Customers")
db.Customers.insertMany([{"cust_id":1, Balance:200, Type:"S"}])
db.Customers.insert({"cust_id":1, Balance:1000, Type:"Z"})
db.Customers.insert({"cust_id":2, Balance:100, Type:"Z"})
db.Customers.insert({"cust_id":2, Balance:1000, Type:"C"})
db.Customers.insert({"cust_id":2, Balance:500, Type:"C"})
db.Customers.insert({"cust_id":2, Balance:50, Type:"S"})
db.Customers.insert({"cust_id":3, Balance:500, Type:"Z"})

db.Customers.aggregate([
  $group: {_id: "$cust_id",
    minAccBal: {$min: "$Balance"},
    maxAccBal: {$max: "$Balance"}}
])

db.Customers.aggregate([
  {$match: {Type: "Z"}},
  {$group: {_id: "$cust_id", TotAccBal: {$sum: "$Balance"}}},
  {$match: {TotAccBal: {$gt: 200}}}
])

// Import / Export
mongoimport --host localhost --db MyDataBase --collection NewStudent2 \
  --type=csv --file /home/bmscecse/Desktop/135.txt --headerline

mongoexport --host localhost --db MyDataBase --collection NewStudent2 \
  --type=json --out /home/bmscecse/Desktop/135.txt

// db.Student.find():
[
```

```
{RollNo:1, Age:21, Cont:9876, email:'antara.de9@gmail.com'},
{RollNo:2, Age:22, Cont:9976, email:'anushka.de9@gmail.com'},
{RollNo:3, Age:21, Cont:5576, email:'anubhav.de9@gmail.com'},
{RollNo:4, Age:20, Cont:4476, email:'pani.de9@gmail.com'},
{RollNo:10, Age:23, Cont:2276, email:'tarun.de9@gmail.com'}
]

// aggregate min/max:
[ {_id:3,minAccBal:500,maxAccBal:500},
  {_id:2,minAccBal:50, maxAccBal:1000},
  {_id:1,minAccBal:200,maxAccBal:1000}]

// filter Z, sum > 200:
[ {_id:3,TotAccBal:500}, {_id:1,TotAccBal:1000}]

2026-01-30T10:22:14.438+0530  connected to: mongodb://localhost/
2026-01-30T10:22:14.780+0530  3 document(s) imported successfully.
2026-01-30T10:22:32.771+0530  connected to: mongodb://localhost/
2026-01-30T10:22:32.780+0530  exported 3 records
```